Lever Test · Tier Pack · Aspire

Learning goal: I can test how the distance from the pivot changes the effort needed to lift a load.

Aspire Route (Book First)

Use the quantitative pattern (5 cm -> 8, 10 cm -> 4, 20 cm -> 2) to explain what happens when effort distance doubles, and state one source of error in a ruler lever.

Help: Every time the effort distance from the pivot doubles, the effort needed __. One source of error is the mass of the __ itself.

Optional Make It · Aspire

Make a Lever Ratio & Limit Card

Show the numerical pattern (doubling distance halves the coin count) and explain why the ruler's own weight can affect the balance.

Word Help

effort distance: the distance from the pivot to where the effort push is applied

load: the weight or mass being lifted by the lever

independent variable: the one thing we change on purpose (the effort distance from the pivot)

pattern: how one measurement changes when we change another

Exit Check

Which effort distance needed the fewest coins: 5 cm or 20 cm?

What happens to the effort needed as you push further from the pivot?

Name two things we kept the same to make the lever test fair.